

CSE5DMI Data Mining

Week 08

ASSOCIATION ANALYSIS: BASIC CONCEPTS AND ALGORITHMS

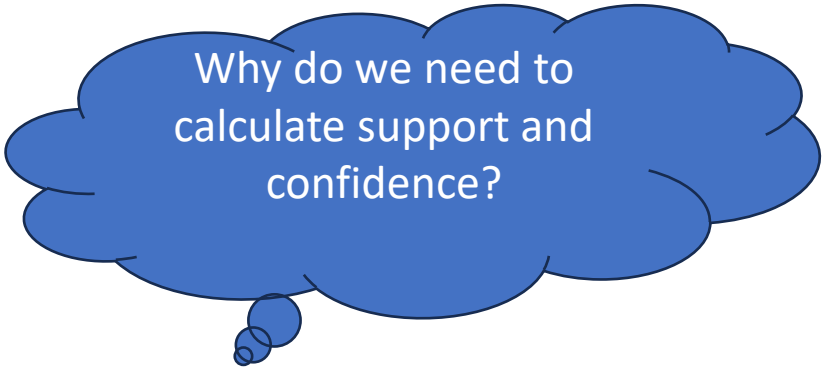
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REVIEW – WEEK 7

- Rule (X->Y) evaluation metrics
  - Support: fraction of transactions that contain both X and Y
  - Confidence: measures how often items in Y appear in transactions that contain X
- Mining association rules: two-step approach
  - Frequent itemset generation

    - An itemset whose support is >= a defined threshold
  - Rule generation (this week)

REVIEW – WEEK 7



Support is an important measure because a rule that has very low support might occur simply by chance.

From business perspective, a low support rule is unlikely to be interesting because it might not make profit by promoting items that customers seldom buy together.

we are interested in finding rules whose support is greater than a user-defined threshold.

Confidence measures the reliability of the inference made by a rule. The higher the confidence, the more likely it is for Y to be present in transactions that contain X

Confidence also provides an estimate of the conditional probability of Y given X.

Note: The inference made by an association rule *does not necessarily imply relationship between cause and effect*. Instead, it can sometimes suggest a strong co-occurrence relationship between items in the X and consequent of the rule.

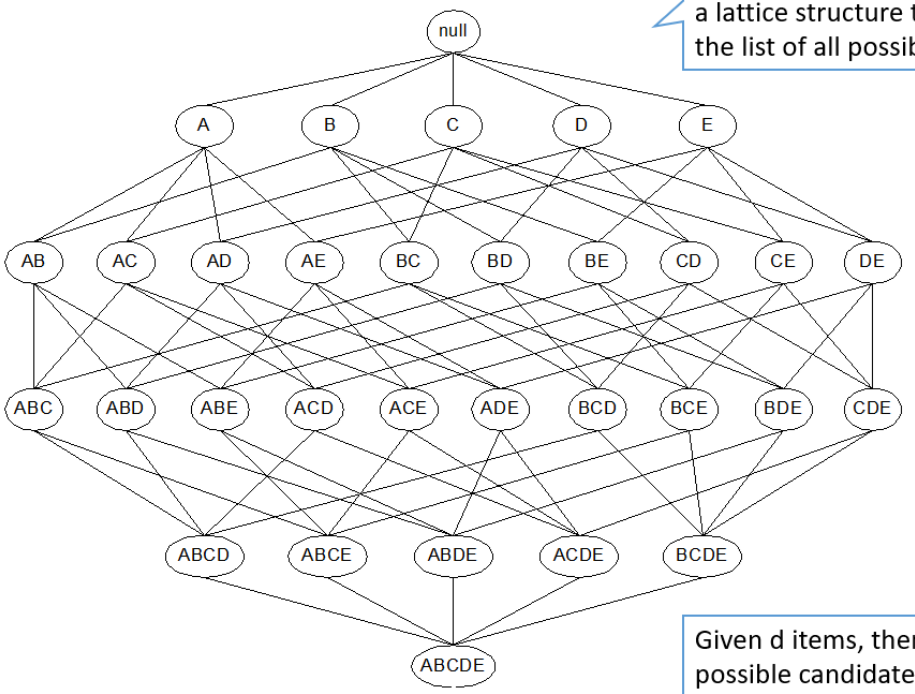
REVIEW – WEEK 7

- Mining association rules: two-step approach
  - Frequent itemset generation

    - An itemset whose support is >= a defined threshold
  - Rule generation (this week)

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Frequent Itemset Generation

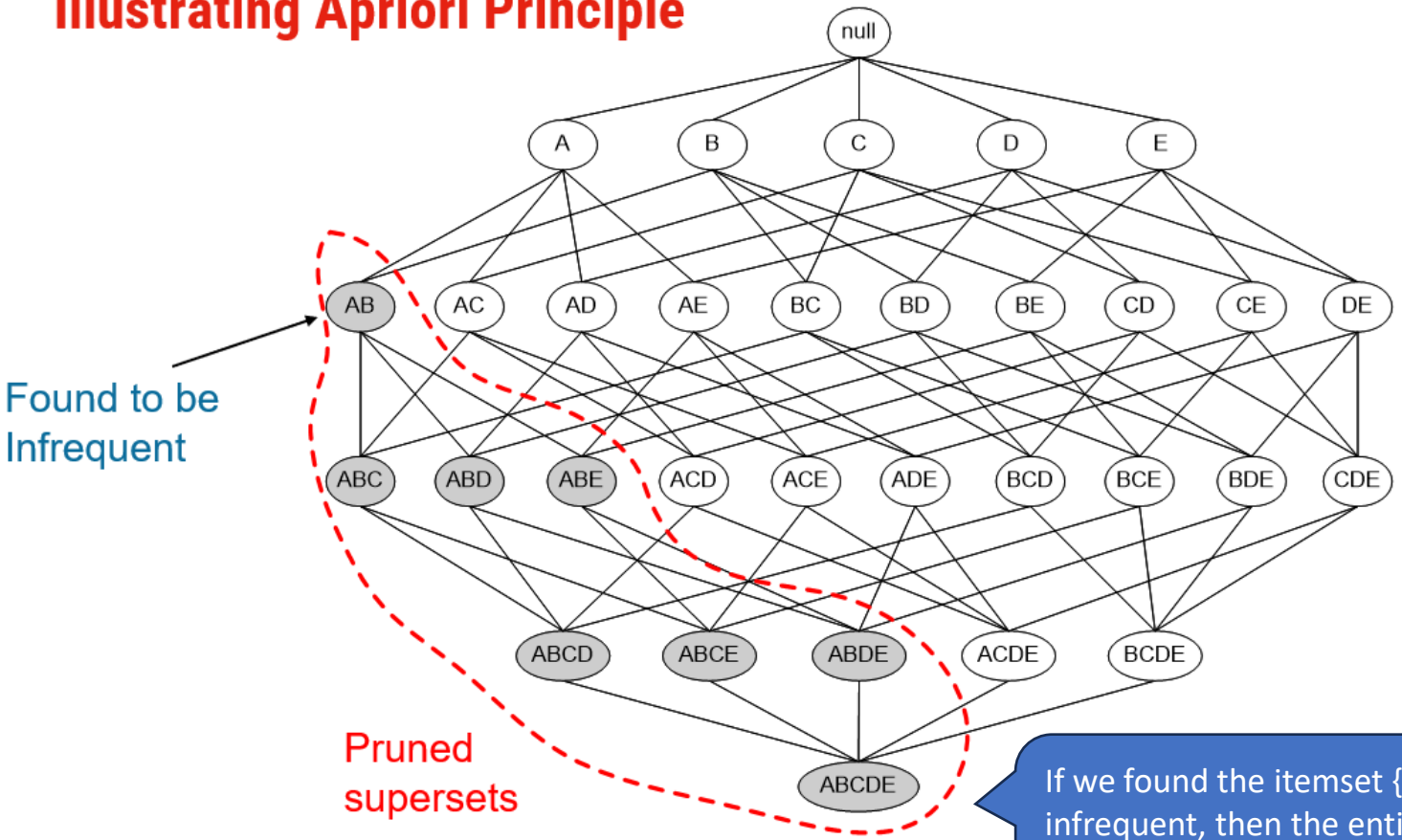


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REVIEW – WEEK 7

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Illustrating Apriori Principle

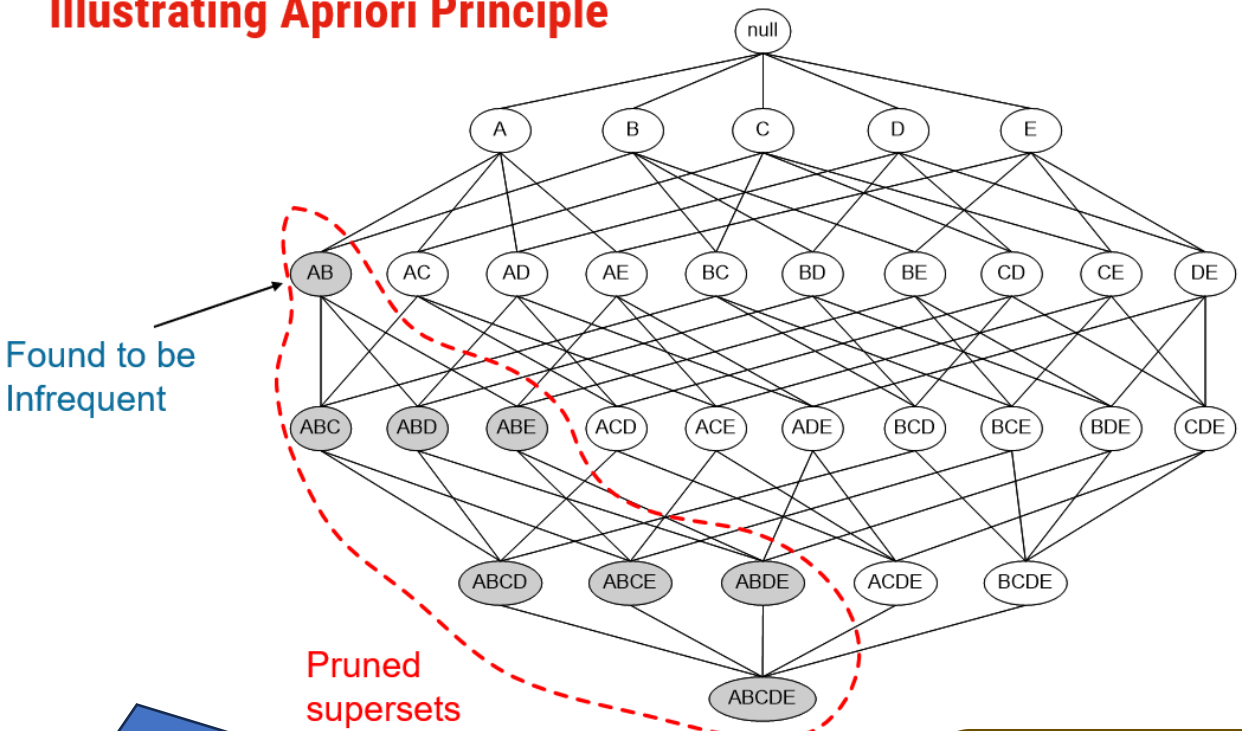


If we found the itemset {a,b} is infrequent, then the entire subgraph containing the supersets of {a,b} can be pruned immediately.

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Illustrating Apriori Principle



If we found the itemset {a,b} is infrequent, then the entire subgraph containing the supersets of {a,b} can be pruned immediately.

That's because: Support of an itemset never exceeds the support of its subsets

REVIEW – WEEK 7

That's because: Support of an itemset never exceeds the support of its subsets

Transaction ID	Items Bought
T1	{Milk, Bread, Eggs}
T2	{Milk, Bread}
T3	{Milk, Diaper, Beer, Eggs}
T4	{Bread, Diaper, Milk, Beer}
T5	{Bread, Milk, Diaper}

Support of single items

- **Milk** appears in T1, T2, T3, T4, T5 → **5 times** → support({Milk}) = 5/5 = **1.0**
- **Bread** appears in T1, T2, T4, T5 → **4 times** → support({Bread}) = 4/5 = **0.8**

Support of a 2-itemset

{Milk, Bread} appears in T1, T2, T4, T5 → **4 times** → support({Milk, Bread}) = 4/5 = **0.8**

Notice: support({Milk, Bread}) ≤ support({Milk}) (0.8 ≤ 1.0).

Support of a 3-itemset

{Milk, Bread, Eggs} appears only in T1 → **1 time** → support = 1/5 = **0.2**

Again: support({Milk, Bread, Eggs}) ≤ support({Milk, Bread}) (0.2 ≤ 0.8).

Observation

support({Milk}) = 1.0

support({Milk, Bread}) = 0.8

support({Milk, Bread, Eggs}) = 0.2

As the itemset **grows larger**, its support **never increases** — it either stays the same or decreases.

This is the **anti-monotone property** (and why the Apriori principle works).

WEEK 8 TOPICS

- Rule generation
- Rule pattern evaluation
    - Interestingness measure
    - Properties of measures

Rule Generation

- Given a frequent itemset L, find all non-empty subsets $f \subset L$ such that $f \rightarrow L - f$ satisfies the minimum confidence requirement
  - If {A,B,C,D} is a frequent itemset, candidate rules:

$ABC \rightarrow D$,	$ABD \rightarrow C$,	$ACD \rightarrow B$,	$BCD \rightarrow A$,
$A \rightarrow BCD$,	$B \rightarrow ACD$,	$C \rightarrow ABD$,	$D \rightarrow ABC$
$AB \rightarrow CD$,	$AC \rightarrow BD$,	$AD \rightarrow BC$,	$BC \rightarrow AD$,
$BD \rightarrow AC$,	$CD \rightarrow AB$,		

- If $|L| = k$, then there are $2^k - 2$ candidate association rules (ignoring $L \rightarrow \varnothing$ and $\varnothing \rightarrow L$)

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Rule Generation

Confidence of Rule X ->L-X: L/X

How to efficiently generate rules from frequent itemsets?

- In general, confidence **does not** have an anti-monotone property

c(ABC →D) can be larger or smaller than c(AB →D)

Why confidence does **not** have an anti-monotone property? Example - 1

TID	Items
1	A, B, C, D
2	A, B, D
3	A, B, C
4	A, C, D
5	B, C, D

Rule 1: $ABC \rightarrow D$
- Support(ABCD) = 2 (TIDs 1, 4)
- Support(ABC) = 2 (TIDs 1, 3)
- Confidence = 2/2 = **1.0**

Rule 2: $AB \rightarrow D$
- Support(ABD) = 2 (TIDs 1, 2)
- Support(AB) = 3 (TIDs 1, 2, 3)
- Confidence = 2/3 = **0.67**

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Rule Generation

Confidence of Rule X ->L-X: L/X

How to efficiently generate rules from frequent itemsets?

- In general, confidence **does not** have an anti-monotone property

c(ABC →D) can be larger or smaller than c(AB →D)

Why confidence does **not** have an anti-monotone property? Example - 2

TID	Items
1	A, B, C, D
2	A, B, C
3	A, B, D
4	A, B, D
5	A, B, D

Rule 1: $ABC \rightarrow D$

•c=support(ABC)/support(ABCD)=2/1=0.5

Rule 2: $AB \rightarrow D$

•c=support(AB)/support(ABD)=5/4=0.8

- When you add more items to the LHS (from AB to ABC), the denominator (support of LHS) usually shrinks.
- Depending on how many of those cases also contain D, confidence can go up or down.
- That's why confidence is not anti-monotone in general with respect to LHS size.

Rule Generation

How to efficiently generate rules from frequent itemsets?

- In general, confidence **does not** have an anti-monotone property

  c(ABC $\rightarrow$D) can be larger or smaller than c(AB $\rightarrow$D)

- But confidence of rules generated from the same itemset has **an anti-monotone** property

  - E.g., Suppose {A,B,C,D} is a frequent 4-itemset:

$$c(ABC \rightarrow D) \geq c(AB \rightarrow CD) \geq c(A \rightarrow BCD)$$

Confidence of Rule X ->L-X: L/X

  - Confidence is anti-monotone w.r.t. number of items on the RHS of the rule

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Rule Generation

Confidence of Rule X ->L-X: L/X

- Why confidence of rules generated from the same itemset has **an anti-monotone** property?

TID	Items
1	A, B, C, D
2	A, B, C, D
3	A, B, C
4	A, B, D
5	A, C, D
6	B, C, D

Support({ABCD}) = 2 (TIDs 1, 2)
Support({ABC}) = 3 (TIDs 1, 2, 3)
Support({AB}) = 4 (TIDs 1, 2, 3, 4)
Support({A}) = 5 (TIDs 1–5)

Rule 1: $ABC \rightarrow D$
c=support(ABCD)/support(ABC)=2/3≈0.67
Rule 2: $AB \rightarrow CD$
c=support(ABCD)/support(AB)=2/4=0.50
Rule 3: $A \rightarrow BCD$
c=support(ABCD)/support(A)=2/5=0.40

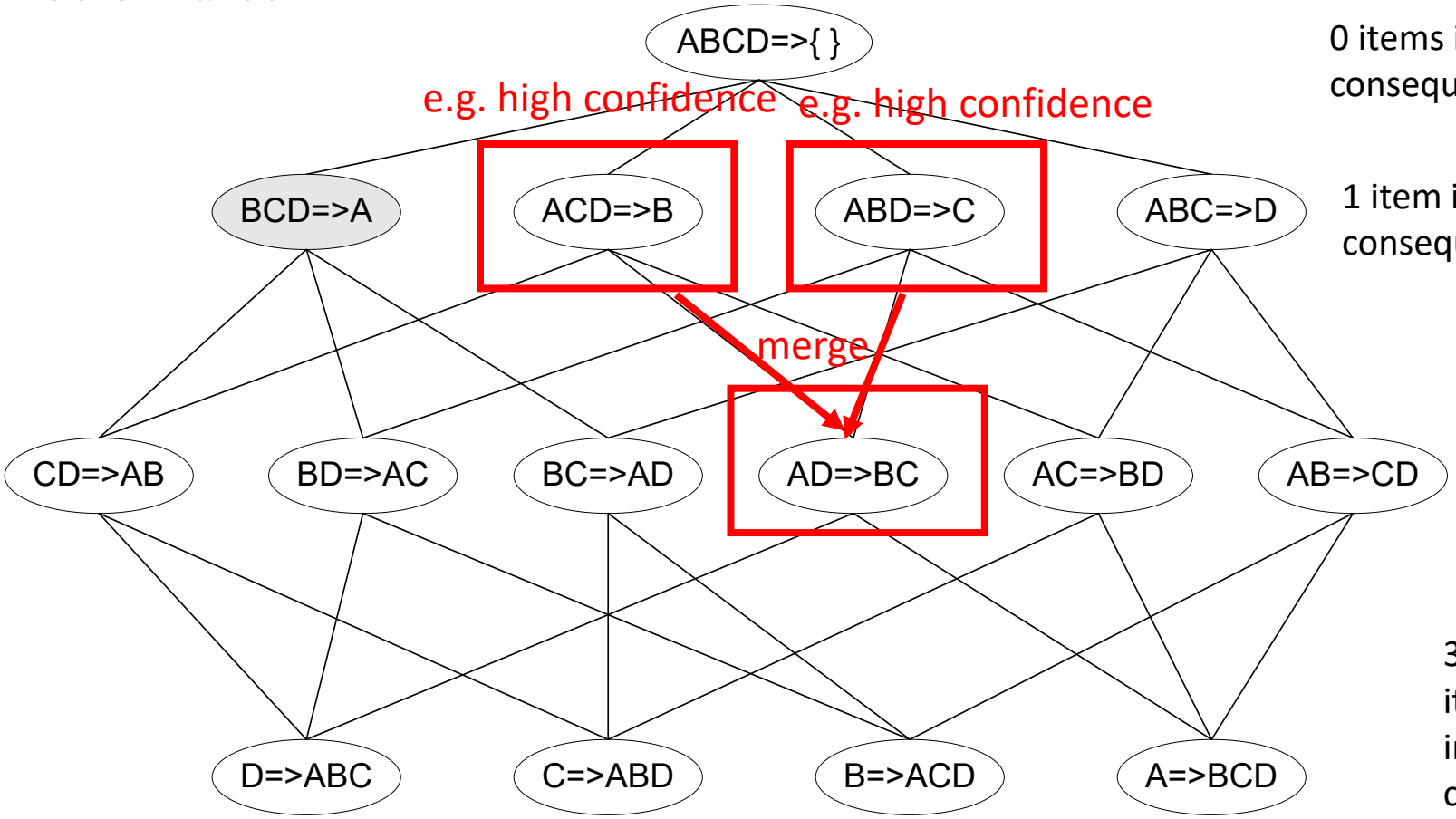
anti-monotone property observed !

$$c(ABC \rightarrow D) = 0.67 \geq c(AB \rightarrow CD) = 0.50 \geq c(A \rightarrow BCD) = 0.40$$

As RHS grows, **confidence decreases or stays equal**.

Rule Generation for Apriori Algorithm

Lattice of rules

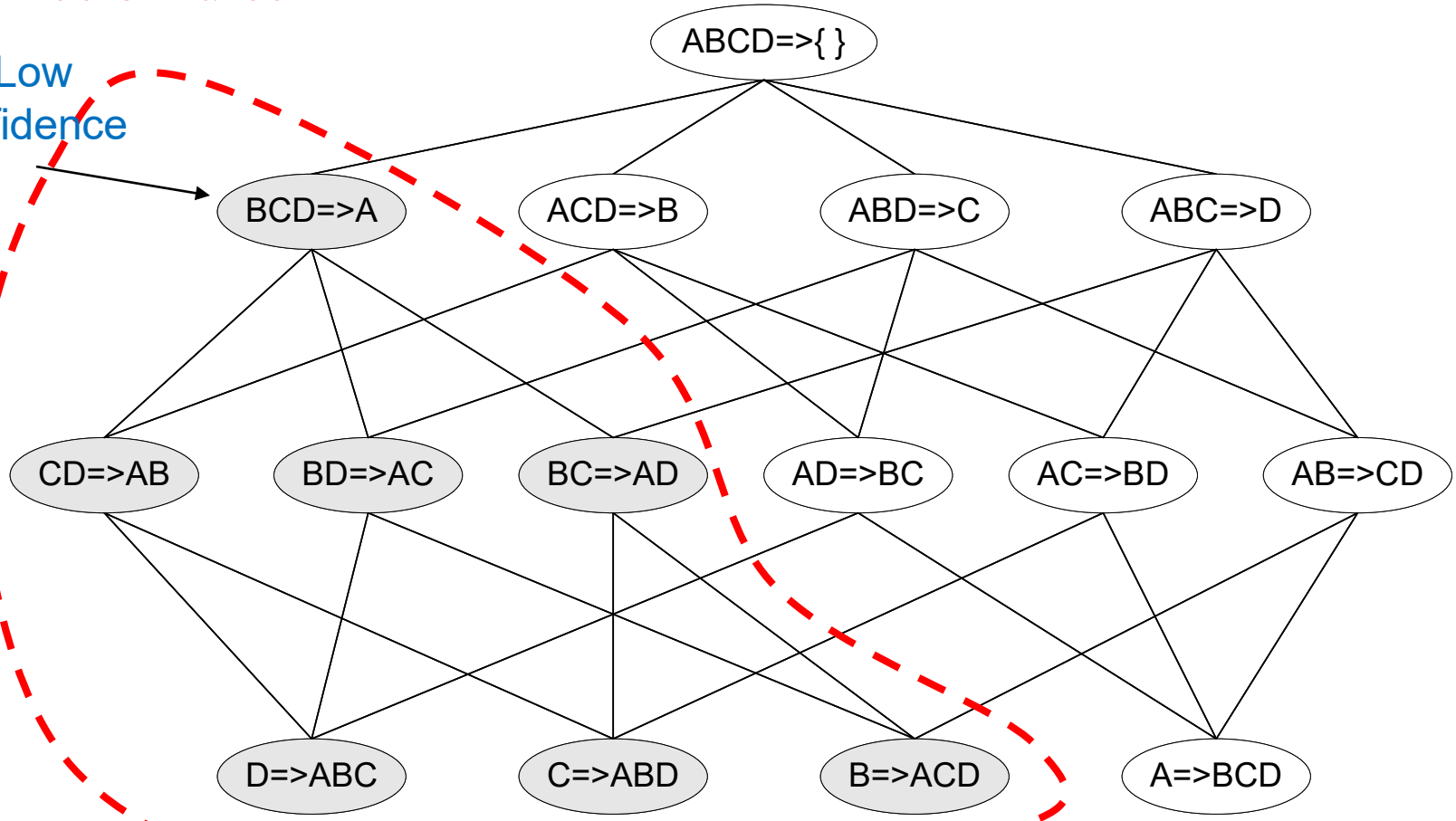


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Rule Generation for Apriori Algorithm

Lattice of rules

e.g. Low Confidence Rule



Sub-graph can be pruned according **anti-monotone** property

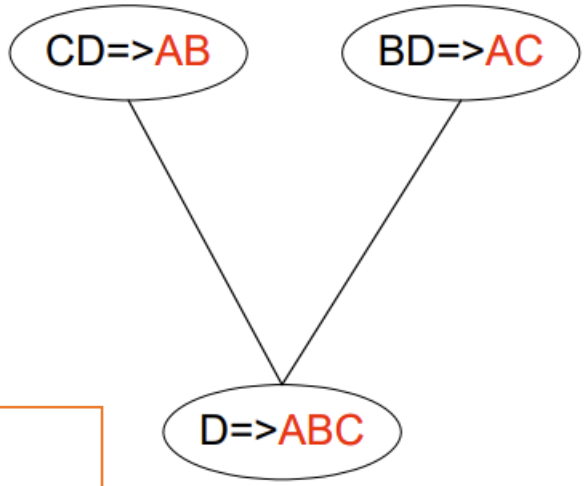
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Rule Generation for Apriori Algorithm

Candidate rule is generated by merging two rules that share the same prefix in the rule consequent

join(CD=>AB,BD=>AC) would produce the candidate rule D => ABC

Prune rule D=>ABC if its subset AD=>BC does not have high confidence



Summary:
- start with rules with only one item in the rule consequence.
- Then generate rules with two items in the rule consequence by merging the consequents of two high confidence rules.
- Next, for the level with 3 items in rule consequence, we merge two rules that share the same prefix in the 2nd level.

During this merging process, if we find that a candidate rule is a low confidence rule, we can prune all its superset immediately.

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Effect of support distribution

How to set the appropriate *minsup* threshold?

- If *minsup* is set too high, we could miss itemsets involving interesting rare items (e.g., expensive products)
- If *minsup* is set too *low*, it is computationally expensive and the number of itemsets is very large

Using a single minimum support threshold may not be effective

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Pattern evaluation

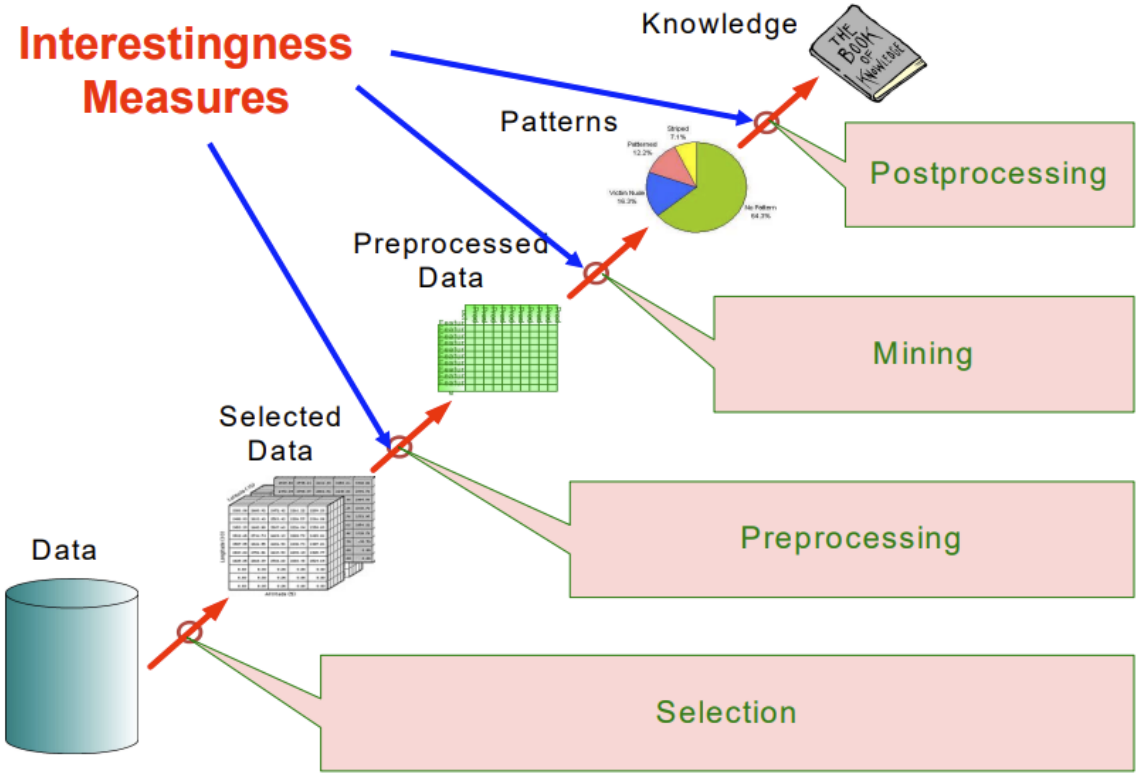
Association rule algorithms tend to produce too many rules

- many of them are uninteresting or redundant
- Redundant if {A,B,C} -> {D} and {A,B} -> {D} have same support & confidence

Interestingness measures can be used to prune/rank the derived patterns

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Application of interestingness measure



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Computing Interestingness Measure

- Given $X \rightarrow Y$, information needed to compute interestingness can be obtained from a contingency table

Contingency table for $X \rightarrow Y$

	Y	$\overline{Y}$	
X	f_{11}	f_{10}	f_{1+}
$\overline{X}$	f_{01}	f_{00}	f_{o+}
	f_{+1}	f_{+0}	N

Used to define various measures

- support, confidence, Gini, entropy, etc.

For example:
f_{11} is the number of times *X* and *Y* appear together in the same transaction
f_{01} is the number of transactions that contain *Y* but not *X*.

For example:
f_{+1} : Support count for Y
f_{1+} : Support count for X

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Discussion

- Suppose we are interested in analyzing the relations between people who drink tea and coffee.
- We collect information about beverage preference among a group of people
- Their responses are summarized by a contingency table.

Customers	Tea	Coffee	…
C1	0	1	…
C2	1	0	…
C3	1	1	…
C4	1	0	…
…			

	Coffee	$\overline{\text{Coffee}}$	
Tea	150	50	200
$\overline{\text{Tea}}$	650	150	800
	800	200	1000

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Discussion

Customers	Tea	Coffee	…
C1	0	1	…
C2	1	0	…
C3	1	1	…
C4	1	0	…
…			

	Coffee	$\overline{\text{Coffee}}$	
Tea	150	50	200
$\overline{\text{Tea}}$	650	150	800
	800	200	1000

Association Rule: Tea → Coffee

Fraction (drink both tea and coffee) = 150/1000 = 15%

Support of Rule: Tea → Coffee = (tea, coffee) / total transactions = 15%

Confidence: (tea, coffee) / (tea) = 150/200 = 75 % # high

Confidence > 50%, meaning people who drink tea are more likely to drink coffee than not drink coffee

So rule seems reasonable

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Discussion

	Coffee	$\overline{\text{Coffee}}$	
Tea	150	50	200
$\overline{\text{Tea}}$	650	150	800
	800	200	1000

Association Rule: Tea $\rightarrow$ Coffee

Thus, the rule tea - > coffee is misleading even though the confidence value is high.

Confidence= P(Coffee|Tea) = 150/200 = 0.75

but P (Coffee) = 0.8, which means knowing that a person drinks tea reduces the probability that the person drinks coffee!

$\Rightarrow$ Note that P(Coffee|$\overline{\text{Tea}}$) = 650/800 = 0.8125

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Discussion – another case

Customers	Tea	Honey	…
C1	0	1	…
C2	1	0	…
C3	1	1	…
C4	1	0	…
…			

	$Honey$	$\overline{Honey}$	
Tea	100	100	200
$\overline{Tea}$	20	780	800
	120	880	1000

Association Rule: Tea → Honey

Confidence of the rule = ?

Fraction of people who buy honey, regardless of whether they drink tea = ?

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Discussion – another case

Customers	Tea	Honey	…
C1	0	1	…
C2	1	0	…
C3	1	1	…
C4	1	0	…
…			

	$Honey$	$\overline{Honey}$	
Tea	100	100	200
$\overline{Tea}$	20	780	800
	120	880	1000

Association Rule: Tea → Honey

Confidence ≅ P(Honey|Tea) = 100/200 = 50%

Confidence = 50%, which may mean that drinking tea has little influence whether honey is used or not

So rule seems uninteresting

But P(Honey) = 120/1000 = 12% (hence tea drinkers are far more likely to have honey

This suggests that there is definitely some information in the preference of a person of buying honey given that he/she buys tea.

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Measure for Association Rules

- So, what kind of rules do we really want?
  - Confidence(X $\rightarrow$ Y) should be sufficiently high
    - To ensure that people who buy X will more likely buy Y than not buy Y
  - Confidence(X $\rightarrow$ Y) > support(Y)
    - Otherwise, rule will be misleading because having item X actually reduces the chance of having item Y in the same transaction
    - Is there any measure that capture this constraint?
      - Answer: Yes. There are many of them.

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Statistical Relationship between X and Y

- The criterion

  $$\text{confidence}(X \rightarrow Y) = \text{support}(Y)$$

  is equivalent to:

  - $P(Y|X) = P(Y)$
  - $P(X,Y) = P(X) \times P(Y)$ (X and Y are independent)

  If $P(X,Y) > P(X) \times P(Y)$: X & Y are positively correlated

  If $P(X,Y) < P(X) \times P(Y)$: X & Y are negatively correlated

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Measures that consider statistical dependence

$$Lift = \frac{P(Y \mid X)}{P(Y)}$$

$$Interest = \frac{P(X,Y)}{P(X)P(Y)}$$

lift is used for rules while interest is used for itemsets

$$PS = P(X,Y) - P(X)P(Y)$$

$$\phi - coefficient = \frac{P(X,Y) - P(X)P(Y)}{\sqrt{P(X)[1-P(X)]P(Y)[1-P(Y)]}}$$

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possibility table for $X \rightarrow Y$

	Y	$\overline{Y}$	
X	f_{11}	f_{10}	f_{1+}
$\overline{X}$	f_{01}	f_{00}	f_{o+}
	f_{+1}	f_{+0}	$\|T\|$

f_{11}: support of X and Y
f_{10}: support of $\underline{X}$ and $\overline{Y}$
f_{01}: support of $\underline{X}$ and $\underline{Y}$
f_{00}: support of X and Y

$$Lift = \frac{P(Y \mid X)}{P(Y)}$$

$$Interest = \frac{P(X,Y)}{P(X)P(Y)}$$

$$PS = P(X,Y) - P(X)P(Y)$$

$$\phi - coefficient = \frac{P(X,Y) - P(X)P(Y)}{\sqrt{P(X)[1-P(X)]P(Y)[1-P(Y)]}}$$

$P(Y|X) = f_{11} / f_{1+}$

$P(Y) = f_{+1} / |T|$

$P(X) = f_{1+} / |T|$

$P(X,Y) = f_{11} / |T|$

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Example: Lift/Interest

	Coffee	$\overline{\text{Coffee}}$	
Tea	150	50	200
$\overline{\text{Tea}}$	650	150	800
	800	200	1000

Association Rule: Tea $\rightarrow$ Coffee

Confidence= P(Coffee|Tea) = 0.75

but P(Coffee) = 0.8

$\Rightarrow$ Interest = 0.15 / (0.2×0.8) = 0.9375 (< 1, therefore is negatively associated)

So, is it enough to use confidence/Interest for pruning?

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There are lots of measures proposed in the literature

Measure (Symbol)	Definition
Correlation (ϕ)	$\frac{Nf_{11}-f_{1+}f_{+1}}{\sqrt{f_{1+}f_{+1}f_{0+}f_{+0}}}$
Odds ratio (α)	$(f_{11}f_{00})/(f_{10}f_{01})$
Kappa (κ)	$\frac{Nf_{11}+Nf_{00}-f_{1+}f_{+1}-f_{0+}f_{+0}}{N^2-f_{1+}f_{+1}-f_{0+}f_{+0}}$
Interest (I)	$(Nf_{11})/(f_{1+}f_{+1})$
Cosine (IS)	$(f_{11})/(\sqrt{f_{1+}f_{+1}})$
Piatetsky-Shapiro (PS)	$\frac{f_{11}}{N}-\frac{f_{1+}f_{+1}}{N^2}$
Collective strength (S)	$\frac{f_{11}+f_{00}}{f_{1+}f_{+1}+f_{0+}f_{+0}}\times\frac{N-f_{1+}f_{+1}-f_{0+}f_{+0}}{N-f_{11}-f_{00}}$
Jaccard (ζ)	$f_{11}/(f_{1+}+f_{+1}-f_{11})$
All-confidence (h)	$\min\left[\frac{f_{11}}{f_{1+}},\frac{f_{11}}{f_{+1}}\right]$

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Different Measures have Different Properties

Symbol	Measure	Inversion	Null Addition	Scaling
ϕ	ϕ-coefficient	Yes	No	No
α	odds ratio	Yes	No	Yes
κ	Cohen's	Yes	No	No
I	Interest	No	No	No
IS	Cosine	No	Yes	No
PS	Piatetsky-Shapiro's	Yes	No	No
S	Collective strength	Yes	No	No
ζ	Jaccard	No	Yes	No
h	All-confidence	No	Yes	No
s	Support	No	No	No

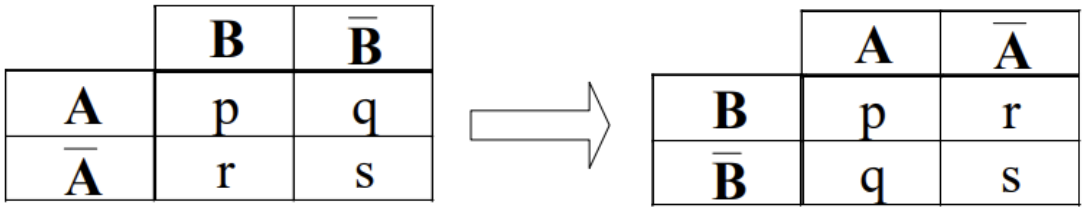
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Properties of a measure

- Symmetric Property
- Inversion Property
- Null Addition Property
- Scaling Property

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Property under Variable Permutation



	B	$\bar{B}$
A	p	q
$\bar{A}$	r	s

	A	$\bar{A}$
B	p	r
$\bar{B}$	q	s

Does M(A,B) = M(B,A)?

Symmetric measures: "Friendship", "no direction"

If Alice is friends with Bob, then Bob is also friends with Alice. It's mutual. Order doesn't matter.

- Measures such as support, interest, collective strength, cosine, Jaccard, etc.

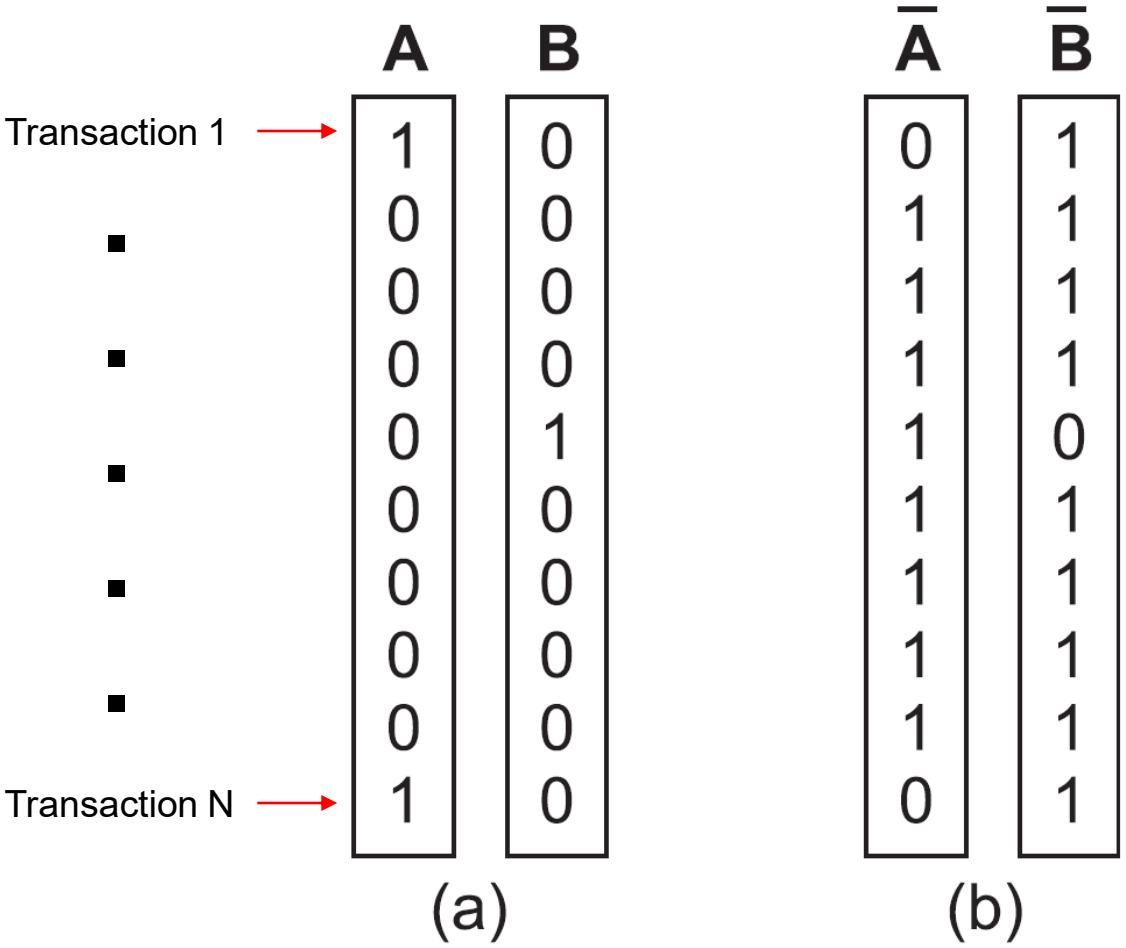
Asymmetric measures: "Teaching", "directional"

If Alice teaches Bob, that does not mean Bob teaches Alice. The direction matters.

- Measures such as confidence, conviction, Laplace, J-measure, etc

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Property under Inversion Operation



The **inversion operation** means:

- Flip every “yes (1)” into “no (0)”,
- and every “no (0)” into “yes (1)”.

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Property under Inversion Operation

Apply inversion (swap 1s and 0s):

- $f_{11} \leftrightarrow f_{00}$, $f_{10} \leftrightarrow f_{01}$.

Correlation (ϕ-coefficient):This value does not change after inversion (it's invariant).

$$\phi = \frac{f_{11}f_{00} - f_{10}f_{01}}{\sqrt{(f_{1.}f_{0.}f_{.1}f_{.0})}}$$

Odds ratio: Also stays the same after inversion. $$OR = \frac{f_{11} \cdot f_{00}}{f_{10} \cdot f_{01}}$$

Interest factor (lift): This one **does change**, because the denominator changes.

$$\text{lift}(A \rightarrow B) = \frac{P(A,B)}{P(A)P(B)}$$

An objective measure *M* is invariant under the inversion operation if its value remains the same when exchanging the frequency counts f11 with f00 and f10 with f01.
Invariant Measures: correlation (ϕ-coefficient), odds ratio, etc.
Measures that do not remain invariant under the inversion operation: the interest factor, etc

Property under Inversion Operation

Apply inversion (swap 1s and 0s):

- $f_{11} \leftrightarrow f_{00}$, $f_{10} \leftrightarrow f_{01}$.

Correlation (ϕ-coefficient) This valu ant).

Odds rati

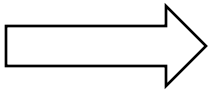
Interest factor (

- Think of inversion like saying:
- "Instead of asking who buys milk and bread together, we ask who does not buy milk and bread together."
- Some measures (like correlation) just care about the overall relationship pattern, so flipping to "not buying" doesn't change the strength.
- Other measures (like lift/interest factor) care about specific probabilities, so flipping changes their values.

An objective measure *M* is invariant under the inversion operation if its value remains the same when exchanging the frequency counts f11 with f00 and f10 with f01.
Invariant Measures: correlation (ϕ-coefficient), odds ratio, etc.
Measures that do not remain invariant under the inversion operation: the interest factor, etc

Property under Null Addition

	B	$\overline{B}$	
A	700	100	800
$\overline{A}$	100	100	200
	800	200	1000



	B	$\overline{B}$	
A	700	100	800
$\overline{A}$	10	1100	1200
	800	1200	2000

Invariant measures:

- cosine, Jaccard, All-confidence, confidence

Non-invariant measures:

- correlation, Interest/Lift, odds ratio, etc

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Property under Null Additi

	B	$\overline{B}$	
A	700	100	800
$\overline{A}$	100	100	200
	800	200	1000

These only care about the "co-occurrence vs. occurrence," not about how many irrelevant transactions are added.

e.g. *"The ratio of people who buy milk and bread among those who buy milk is stable even if we add more people who buy neither."*

Invariant measures:

- cosine, Jaccard, All-c

Non-invariant measures

- correlation, Int

These use the total dataset size or background frequencies in the denominator. Adding nulls changes the ratio.

e.g. "If the supermarket doubles the number of people who buy nothing, the calculated lift of milk→bread changes, even though the real relationship hasn't."

kinds of clever

Property under Row/Column Scaling

Grade-Gender Example (Mosteller, 1968):

	Male	Female	
High	30	20	50
Low	40	10	50
	70	30	100

	Male	Female	
High	60	60	120
Low	80	30	110
	140	90	230

2x 3x

Mosteller:

Underlying association should be independent of the relative number of male and female students in the samples

Odds-Ratio $((f_{11}+f_{00})/(f_{10}+f_{10}))$ has this property

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Property under Row/Column Scaling

Relationship between Mask use and susceptibility to Covid:

	Covid-Positive	Covid-Free	
Mask	20	30	50
No-Mask	40	10	50
	60	40	100

	Covid-Positive	Covid-Free	
Mask	40	300	340
No-Mask	80	100	180
	120	400	520

2x 10x

Mosteller:

Underlying association should be independent of
the relative number of Covid-positive and Covid-free subjects

Odds-Ratio $((f_{11}+f_{00})/(f_{10}+f_{10}))$ has this property

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WEEK 8 SUMMARY

- Rule generation for Apriori algorithm
  - Rule pruning by anti-monotone property
- Rule pattern evaluation
  - Contingency table
  - Calculating interestingness measure
  - Properties of different measures

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Thank you

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